

Hadoop Distribution

- Amazon EMR
 - <https://aws.amazon.com/emr/features/hadoop/>
- Cloudera
 - <http://www.cloudera.com/content/cloudera/en/downloads.html>
- BigInsight
 - <http://www-01.ibm.com/software/data/infosphere/hadoop/products.html>

A Hadoop Cluster

- Hadoop is NOT hardware that you purchase or lease
- Hadoop is actually the name of the software that runs on a cluster
- Both HDFS and YARN work in concert to manage a Hadoop cluster
- Daemon process – software that runs in the background does not require user inputs

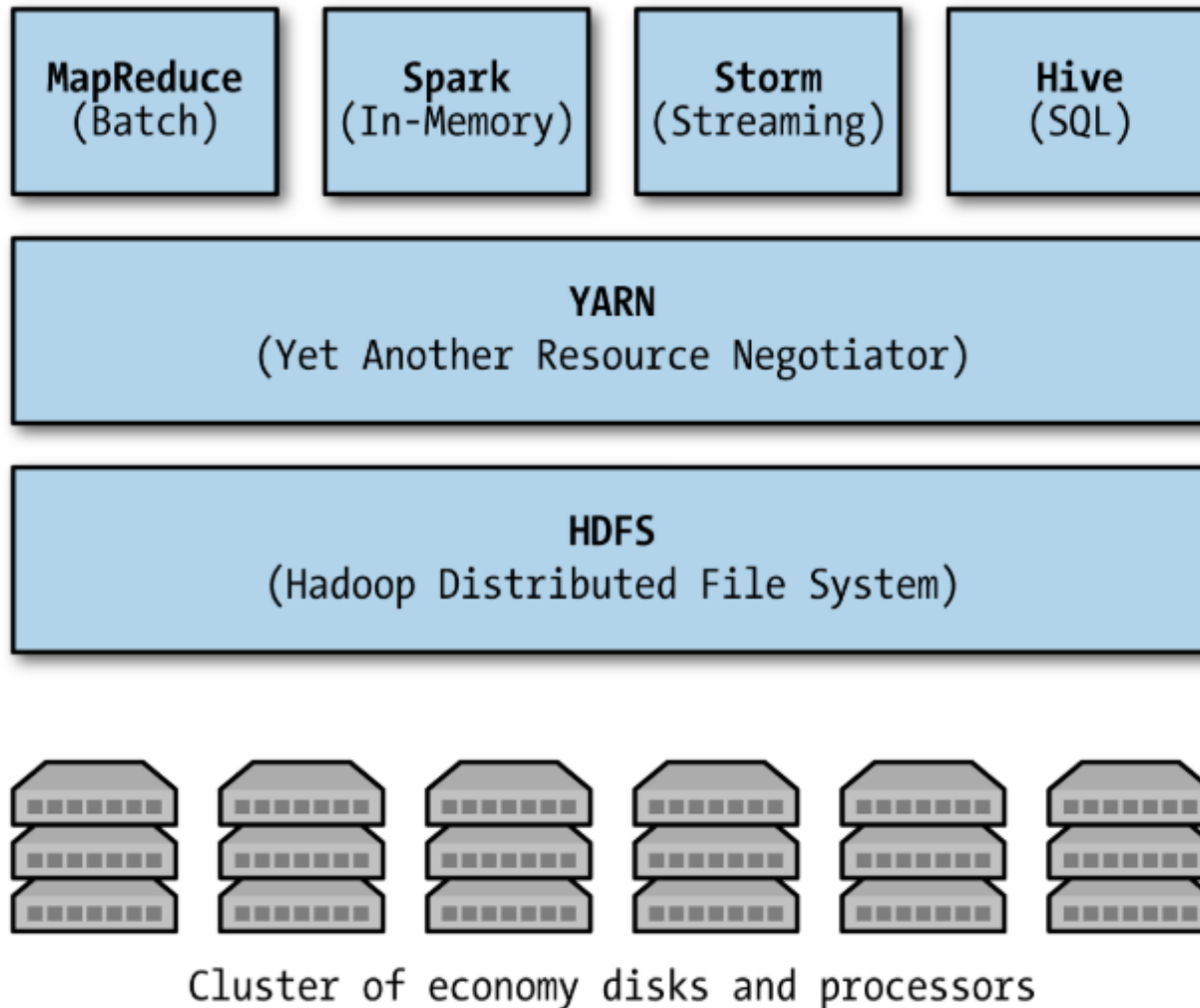
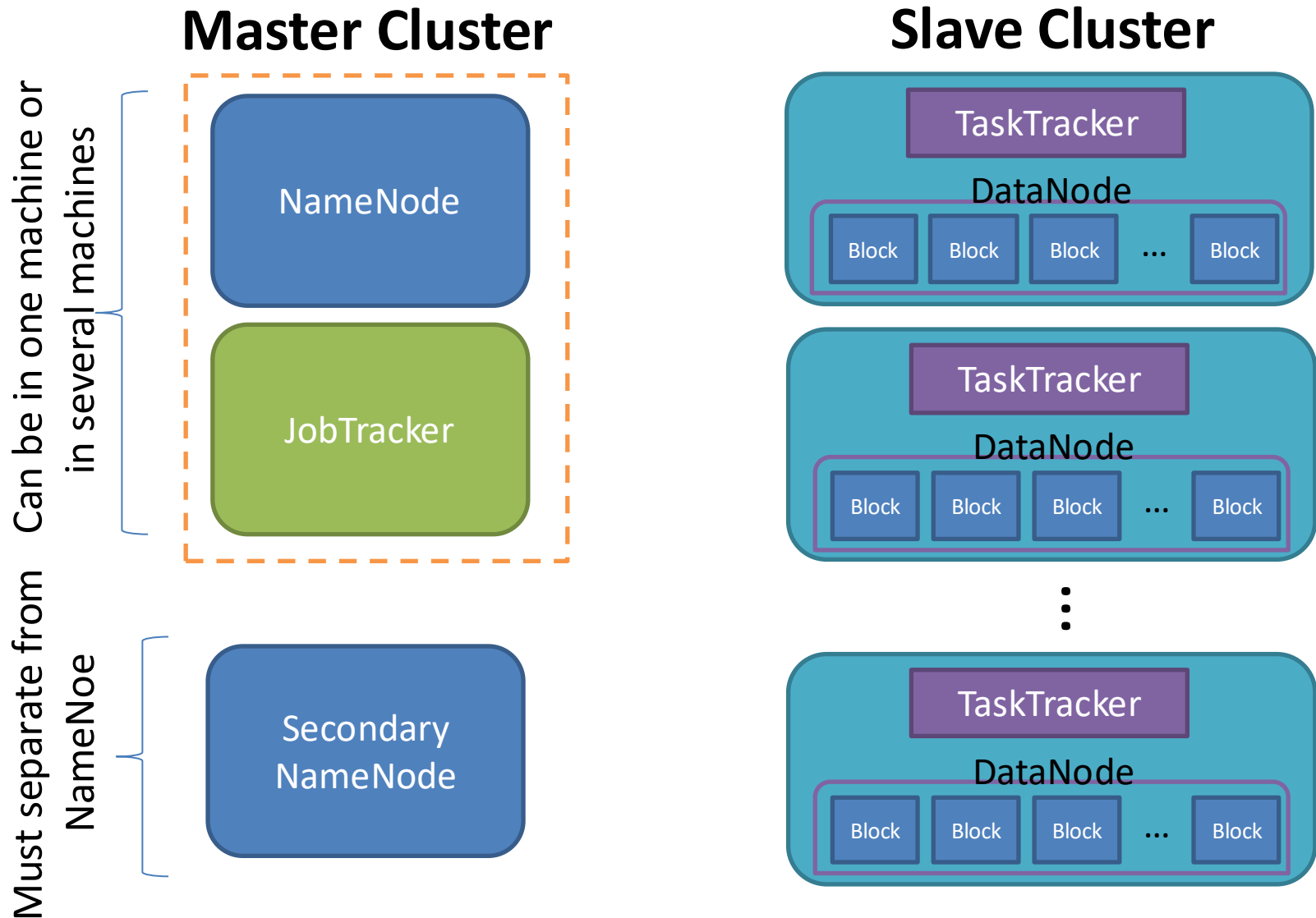


Figure 2-1. Hadoop is made up of HDFS and YARN

Bengfort and Kim 2014

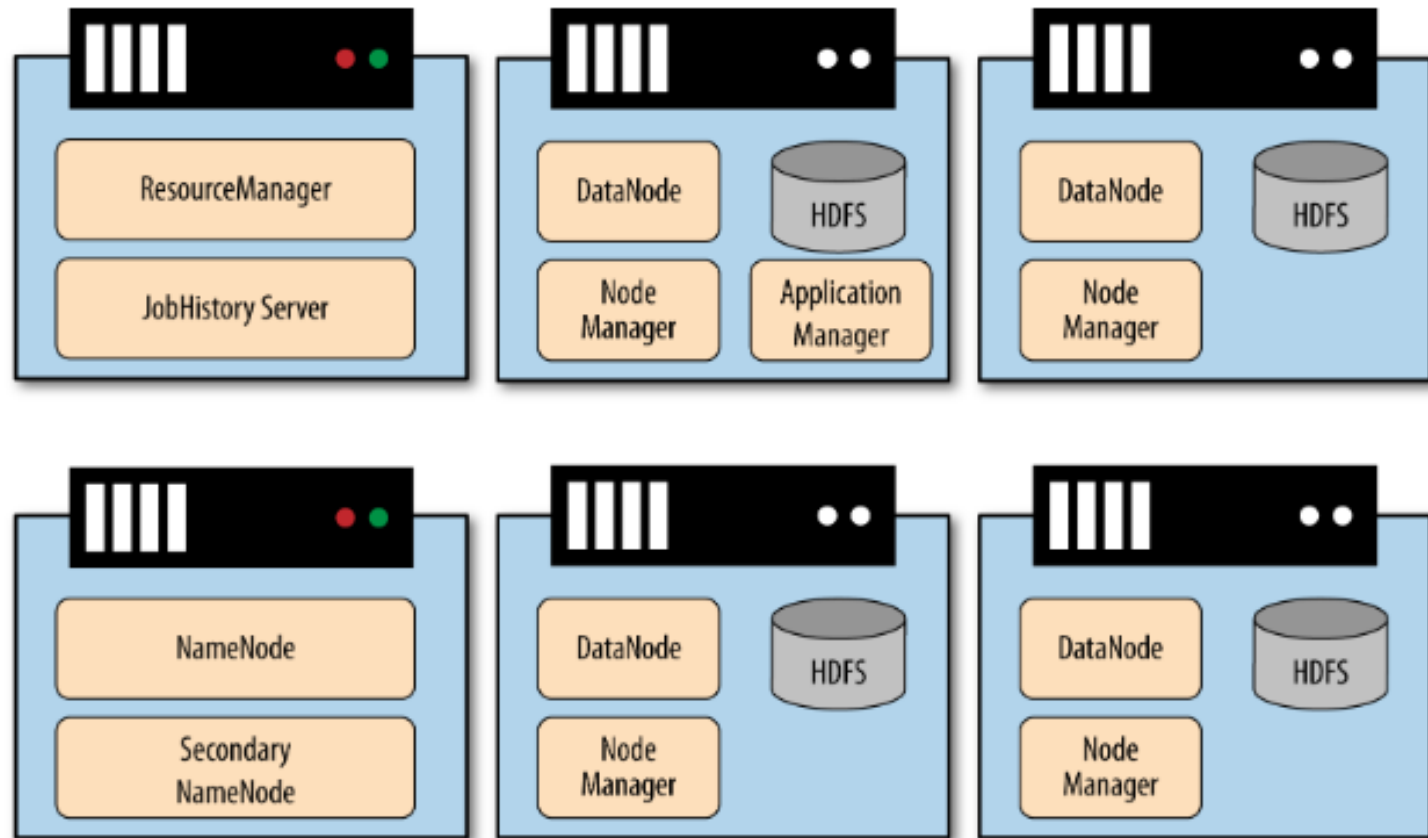
Hadoop Master-Slave Architecture



Your Turn

- What is ZooKeeper in the Hadoop Master and Slave Architecture?
- What are the two main components in ZooKeeper?
- Why was it named this way?

A small Hadoop cluster with two master nodes and four worker nodes



Hadoop Distributed File System

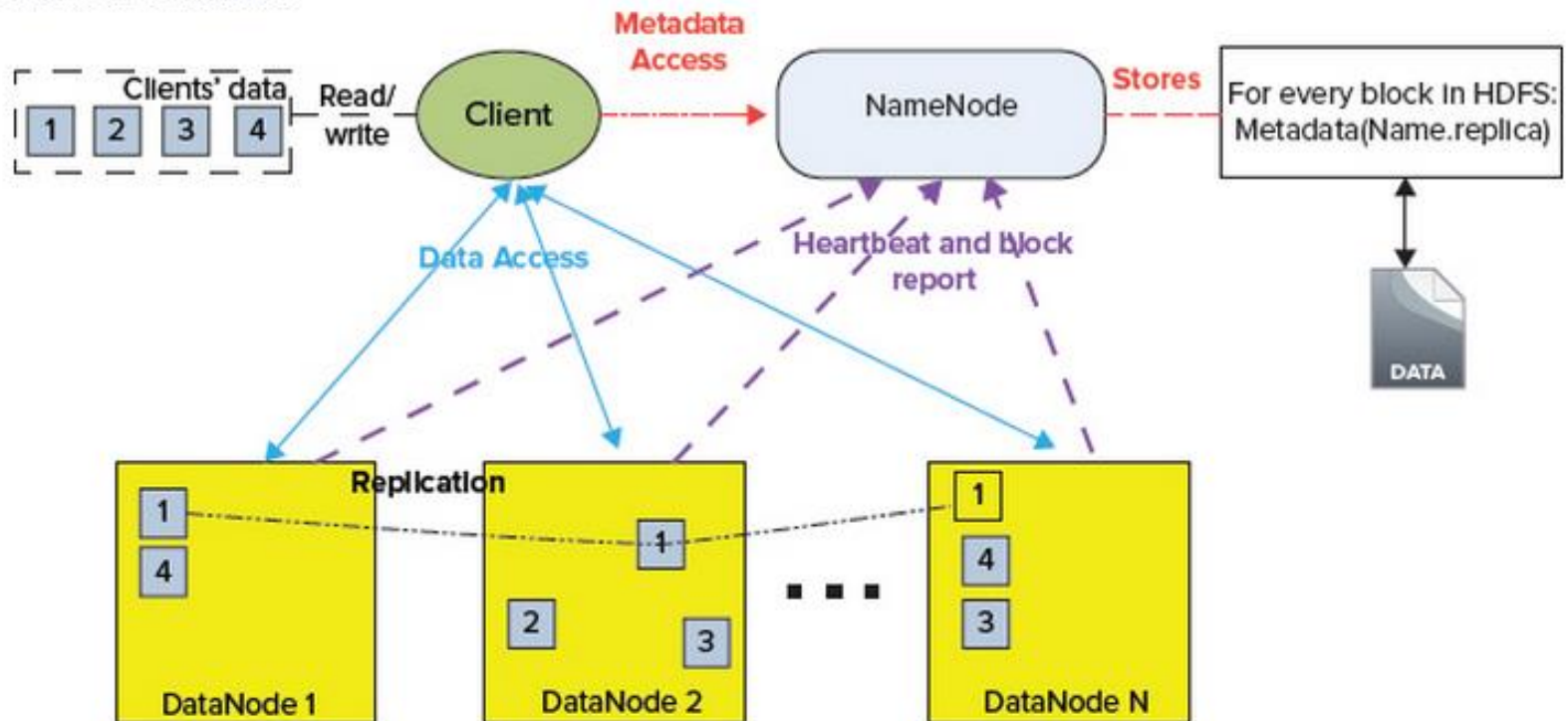
- The HDFS design is based on the design of the Google File System (GFS)
- HDFS is designed to spread the data across a large number of machines.
- HDFS uses data replication to cope with the malfunctioning.
- HDFS is optimized to support high-streaming read performance. This means that if an application is reading from HDFS, it should avoid (or at least minimize) the number of seeks.

Hadoop Distributed File System (cont.)

- HDFS supports only a limited set of operations on files — writes, deletes, appends, and reads, but not updates.
- HDFS does not provide a mechanism for local caching of data. Data should simply be re-read from the source.
- HDFS is implemented as a block-structured file system. The default data block size is 64MB

HDFS Architecture

FIGURE 2-1: HDFS architecture

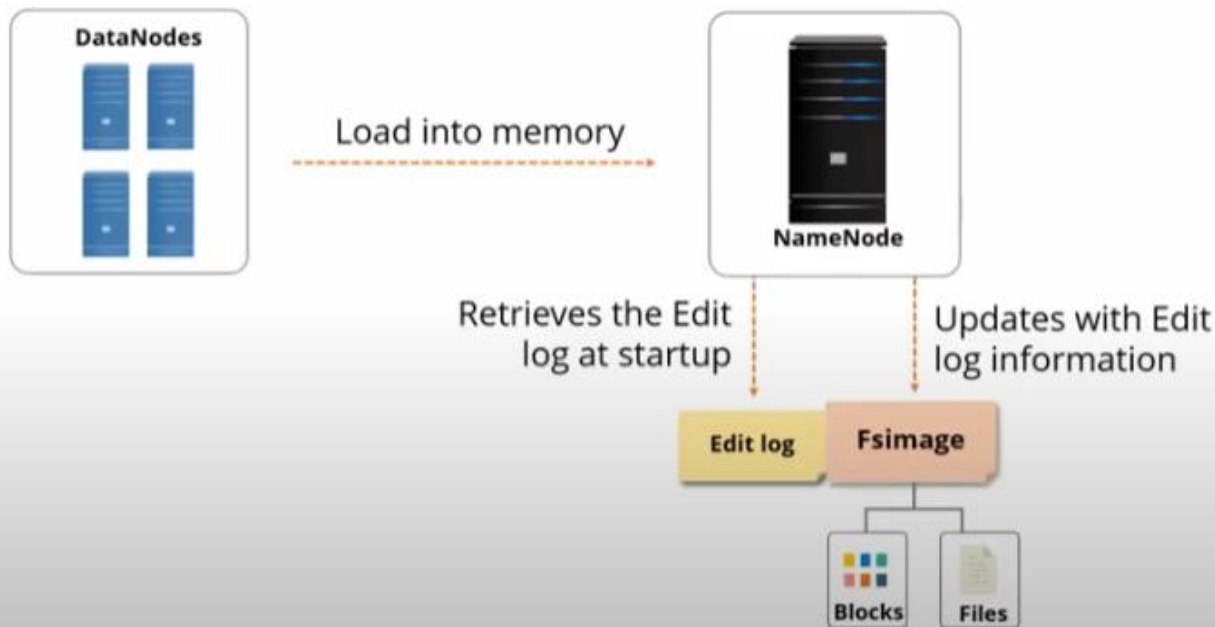


Source: Professional Hadoop Solutions (Ch2)

NameNode Operation

The NameNode maintains two persistent files:

- A transaction log called an Edit Log
- A namespace image called an FsImage



Secondary Namenode

- What if primary namenode failure?
 - losing the NameNode effectively means losing HDFS.
- The Secondary NameNode is not just a “backup NameNode.”
- It serves as a checkpointing mechanism for the primary NameNode.
 - it maintains an image file and an edit log.
 - The image file represents an HDFS metadata state at a point in time, and the edit log is a transactional log of every filesystem metadata change since the image file was created.

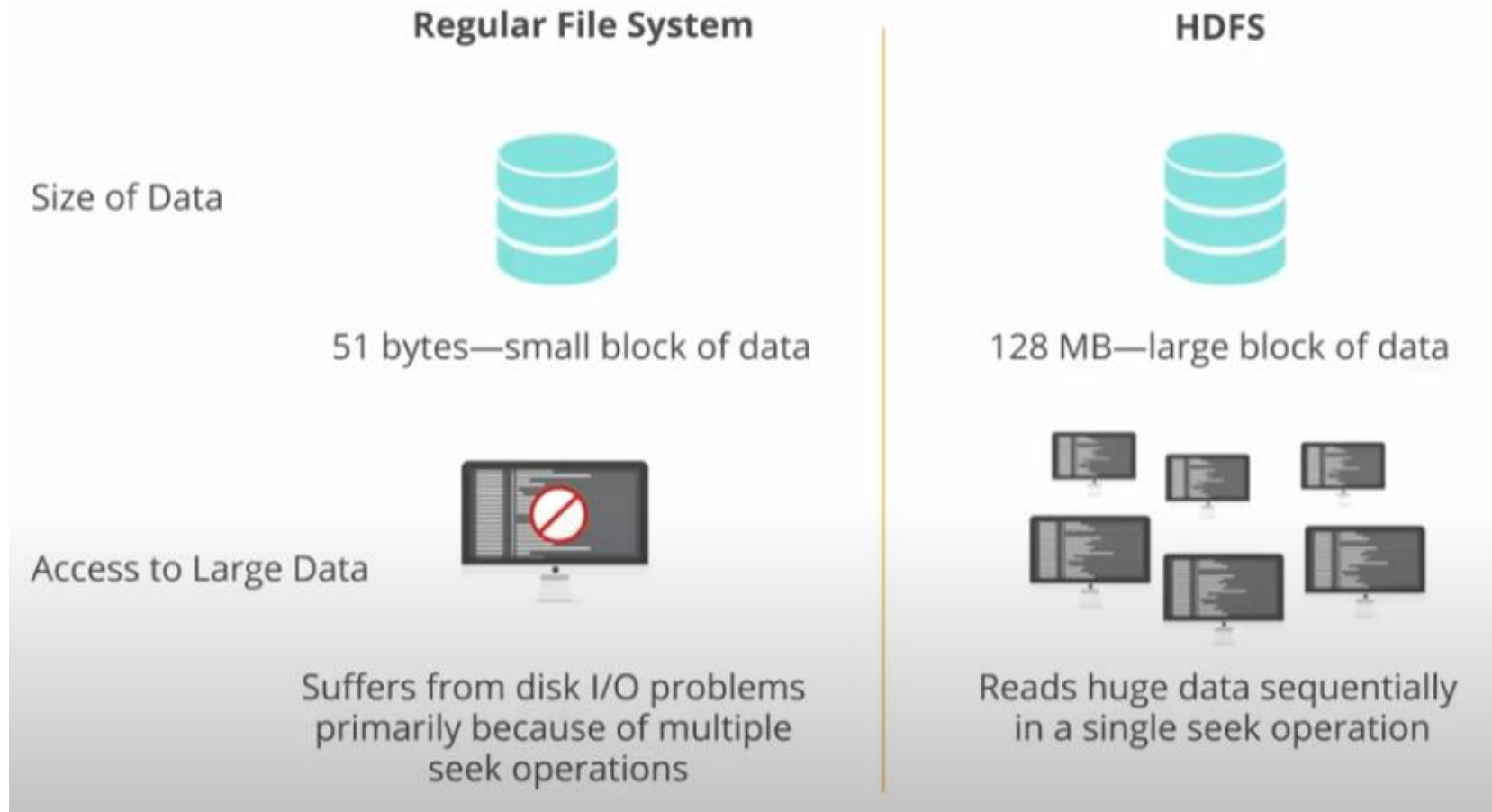
How-to Select the Right Hardware for Your New Hadoop Cluster

- To select the right hardware for a new Hadoop cluster, prioritize factors like storage capacity based on your data volume, sufficient RAM for processing, a robust network for data transfer, and a suitable CPU with multiple cores, while considering the type of workloads and your budget constraints; ensure to choose hardware with redundancy features to handle potential failures and optimize performance across all cluster nodes.

Advantages of HDS

- Cost: zero licensing and support costs
- Speed: Hadoop clusters read or write more than on terabytes of data per second
- Reliability: HDFS copies the data multiple times

Regular File System vs HDFS



Characteristics of HDFS

- Fault Tolerant
- Scalable
- Rack-Aware (40-50 servers per rack)
- Support for heterogeneous clusters
- Build for large datasets

Hadoop File System Commands

- <https://hadoop.apache.org/docs/current2/hadoop-p-project-dist/hadoop-common/FileSystemShell.html>
 - List all files:
 - `hdfs dfs -ls`
 - Copy a file from local machine to HDFS:
 - `hdfs dfs -put <localsrc> <dest>`
 - Copy a file from HDFS to local machine:
 - `hdfs dfs -get <src> <localdest>`
 - Remove file from HDFS:
 - `hdfs dfs -rm <file>`
 - Remove a directory from HDFS:
 - `hdfs dfs -rm -r <folder>`

Tidy Up Your Root Directory

- Create a directory (e.g. project 1)
 - `hdfs dfs -mkdir /user/changs/project1`
- Move a file or **directory** into a **directory**
 - `hdfs dfs -mv out4 project1`
- Copy a file or **directory** into a **directory**
 - `hdfs dfs -cp out3 project1`

Job Run in YARN

- A job run using YARN (see 4:20 – 9:38)
- <https://www.youtube.com/watch?v=oz8TwPgDhc4>

Hive Overview

- Hive description:
 - Hive is a data warehouse infrastructure built on top of Hadoop that provides tools to enable easy data summarization, ad-hoc querying and analysis of large datasets data stored in Hadoop files.
 - It provides a mechanism to put structure on this data and it also provides a simple query language called Hive QL which is based on SQL and which enables users familiar with SQL to query this data.
 - At the same time, this language also allows traditional map/reduce programmers to be able to plug in their custom mappers and reducers to do more sophisticated analysis which may not be supported by the built-in capabilities of the language.

Hive Sources

- Hive Wiki
 - <http://hive.apache.org/>
- Hive QL ref.
 - <https://cwiki.apache.org/confluence/display/Hive/LanguageManual>

Hive Exercise

- Display tables:
 - Show tables;
- Create table
 - CREATE TABLE wordcount (word STRING, freq INT)
ROW FORMAT DELIMITED FIELDS TERMINATED BY
'\t' STORED AS TEXTFILE;
 - Delimited clause to read delimited files
- Description of wordcount table
 - DESCRIBE wordcount;

Hive Exercise (cont.1)

- Load file from HDFS to Hive
 - `LOAD DATA INPATH "word_output" INTO TABLE wordcount;`
- Display the input data:
 - `SELECT * FROM wordcount;`
- Find freq over 2
 - `SELECT * FROM wordcount WHERE freq > 2 SORT BY freq ASC;`
 - `SELECT freq, COUNT(1) AS f2 FROM wordcount GROUP BY freq SORT BY f2 DESC;`

MapReduce

- MapReduce is a programming model for data processing. The model is simple, yet not too simple to express useful programs in
- MapReduce is a framework for executing highly parallelizable and distributable algorithms across huge data sets using a large number of commodity computers.
- MapReduce was introduced to solve large-data computational problems, and is specifically designed to run on commodity hardware. It is based on *divide-and-conquer* principles — the input data sets are split into independent chunks, which are processed by the mappers in parallel.

hadoop overview

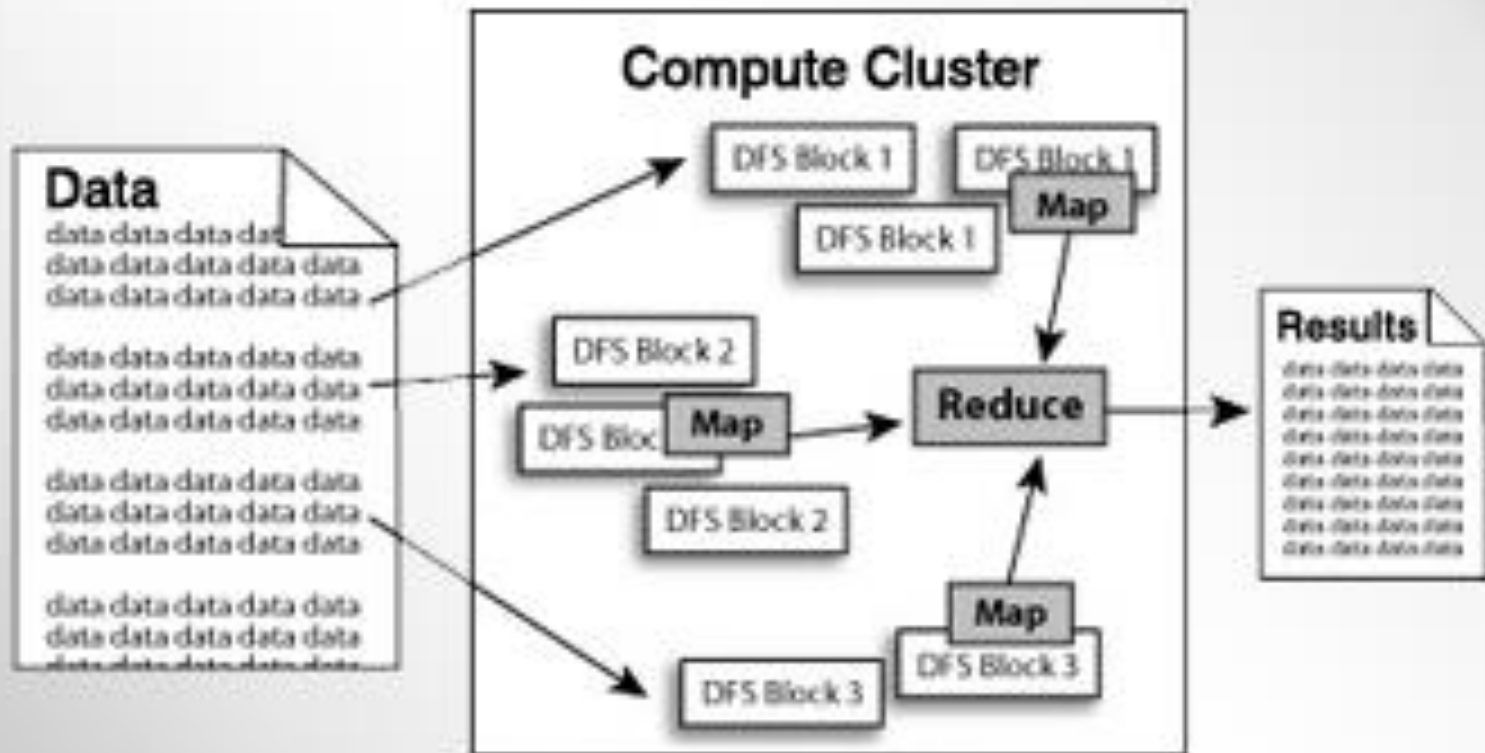


image courtesy of the
Apache Software Foundation

Source: <http://www-01.ibm.com/software/ebusiness/jstart/hadoop/>

How Hadoop Runs a MapReduce Job

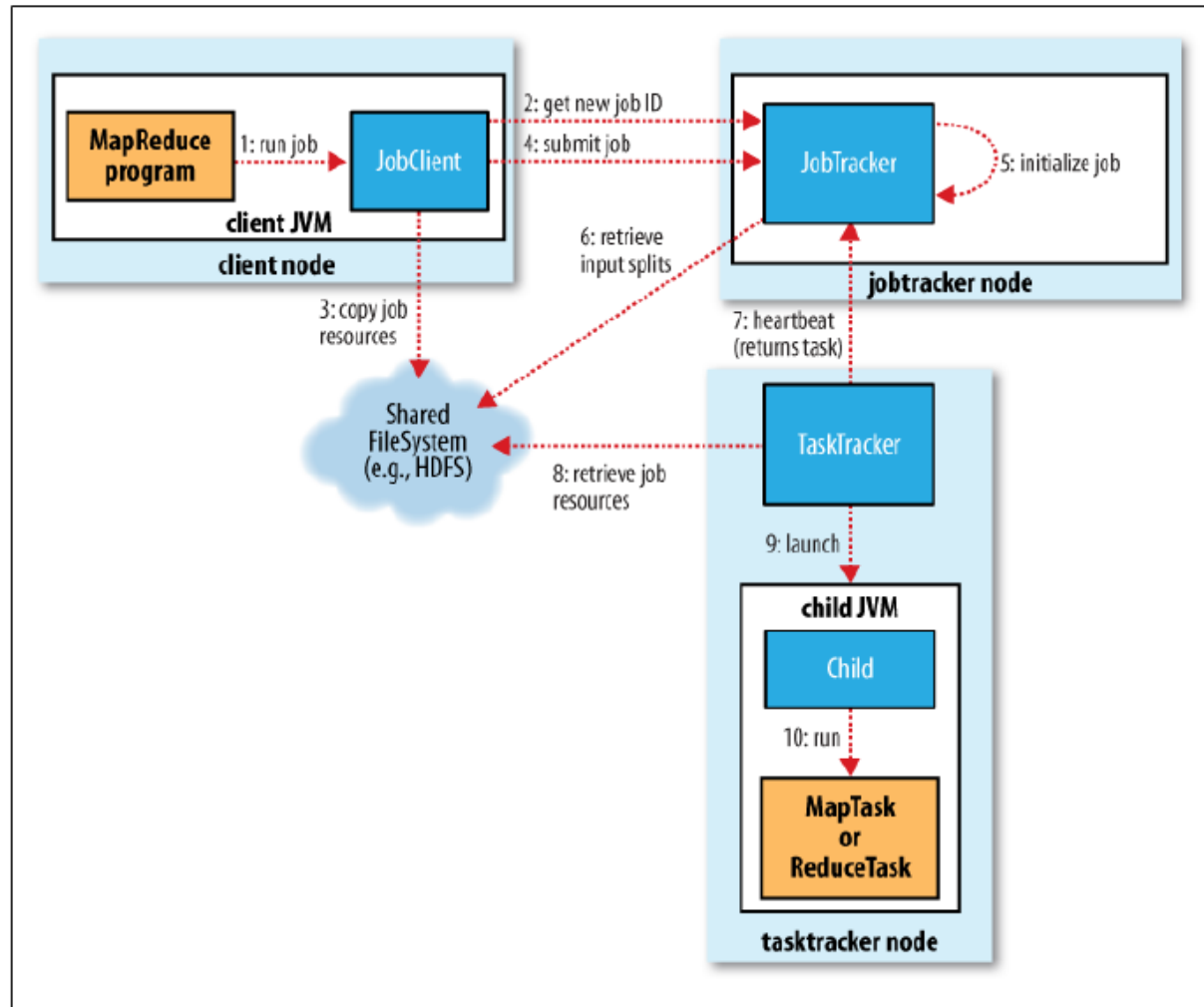


Figure 6-1. How Hadoop runs a MapReduce job using the classic framework

Source: Hadoop: the Definitive guide 3rd (see Ch7 for the YARN version)

MapReduce Hello World Example

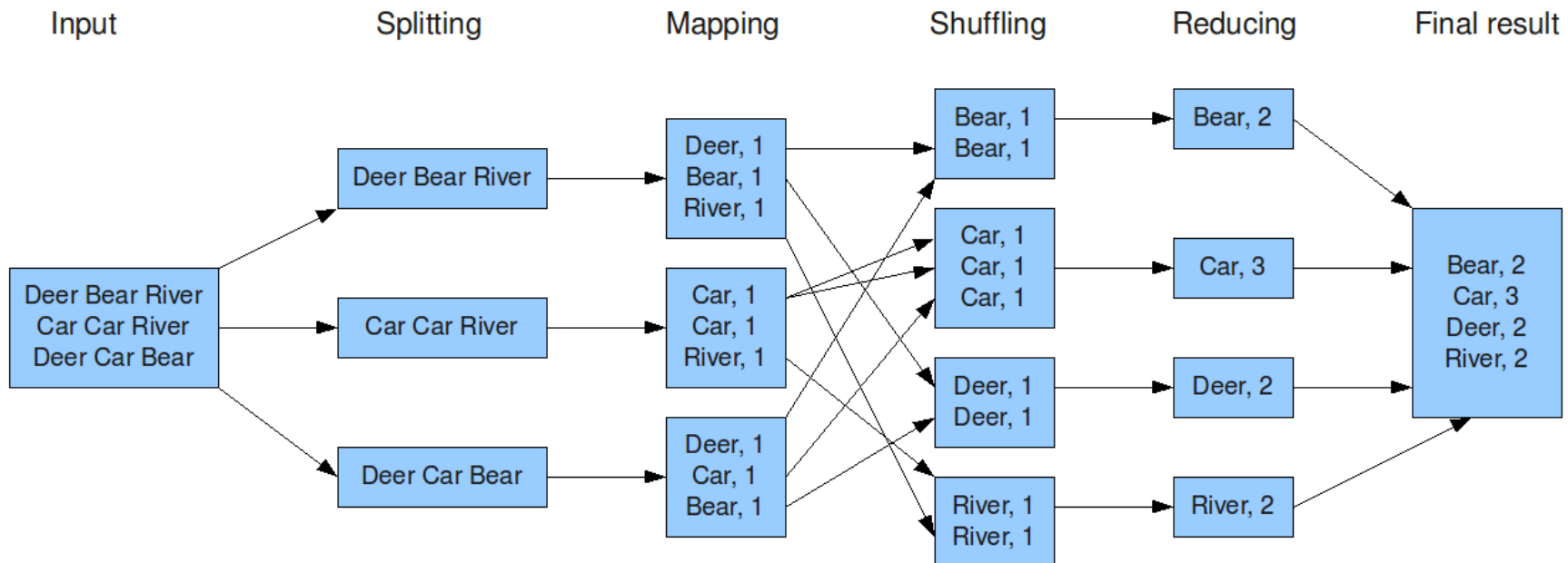
- Create a text file with the following texts:
 - Beer Bear River
 - Car Car River
 - Deer Car Bear
- Put text.txt file to HDFS
- Put the Java codes into HDFS (2nd one):
`$HADOOP_HOME/share/hadoop/mapreduce/hadoop-mapreduce-examples-3.3.0.jar` or
`hadoop-examples.jar`

MapReduce Hello World Example

- Type “**hadoop jar hadoop-examples.jar wordcount text.txt output**” to run wordcount program.
- Once the process is finished, you will see a folder called “output” in your HDFS.
- Type “**hadoop fs -ls output**” to see the output files.
- Type “**hadoop fs -cat output/part-r-00000**” to see the output file.
- Type “**hadoop fs -getmerge output output-local-file**” to move the outcome to local file system

MapReduce Hello World Example

The overall MapReduce word count process



Your Turn

- Use Chat-GPT to find out what other functions are available in the mapreduce java code
- Run a few of the other functions using hadoop jar commands